The unit test

A unit test checks whether a small part of a program, such as a method, works correctly. It compares the result we expect with the actual result. For example, if a method adds 2 and 3, the expected result is 5. Unit tests help programmers find mistakes early and check that their code still works after changes.

JUnit is a tool used to write and run tests in Java. It runs the tests automatically and shows whether they pass or fail. To write a test, we use @Test to identify the test method and an assertion, such as assertEquals, to check the result.

For example, if we already have a Calculator class with an add method, we can write this test:

We can give it 2 and 3 and check that the answer is 5.

In my example, the expected answer is 5. If the method returns 5, the test passes. If it returns a different answer, the test fails.

```
import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.assertEquals;

class CalculatorTest {

    @Test
    void shouldAddTwoNumbers() {
        Calculator calculator = new Calculator();

        int result = calculator.add(2, 3);

        assertEquals(5, result);
    }
}
```

Unit tests usually follow the AAA structure: Arrange, Act, and Assert. First, we prepare the objects and input data. Then, we run the method we want to test. Finally, we check that the actual result matches the expected result

